AGROSPEAR CROPCARE CO., LTD.

FUNGICIDE MANUFACTURING QUALITY CONTROL SYSTEM & DOSSIER

杀菌剂制剂生产全流程质量控制体系与佐证报告规程

Document Code: SOP-QC-FUN-2026
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Version: Rev 4.0 (Global Standard)
Compliance: FAO/WHO Specs & CIPAC Methods
Product Lines: SC, WDG, FS, EW, WP Fungicides

1. END-TO-END FUNGICIDE QA/QC WORKFLOW / 杀菌剂生产品控全生命周期流程

QC Stage / 品控阶段	Key Control Parameters & Protocols / 关键管控项目与规范	Methodology / 检验方法与仪器	Release Criteria / 放行准则
IQC (Incoming QC) **原辅料进厂检验**	**• Technical Material (TC): AI content, enantiomer ratio (e.g., Difenoconazole isomers), relevant toxic impurities (e.g., ETU in Dithiocarbamates), moisture.** • 原药(TC): 有效成分纯度、手性同分异构体比例（如苯醚甲环唑异构体）、限制性有害杂质（如代森类 ETU 乙撑硫脲限度）及水分。 **• Adjuvants & Carriers: Wetting speed, dispersibility index, carrier oil absorption (for WP/WDG), silica moisture.** • 助剂与填料: 润湿速度、分散能力指数、高岭土/白炭黑吸油值与载体含水量。 **• Packaging: Multi-layer co-extruded barrier bottles (EVOH/ HDPE) leak-proof & drop test.** • 包装物料: 多层共挤阻隔瓶（EVOH/HDPE）耐跌落冲击与电磁感应铝箔垫片密闭性测试。	• Chiral HPLC / GC-FID • Karl Fischer Titrator • Drop & Vacuum Leak Tester	100% Passed 不合格严禁入库投料
IPQC (In-Process) **制程过程巡检**	**• Suspension Conc. (SC): Superfine bead milling, monitor crystallization & Ostwald ripening, milling temp < 35°C, fineness D90 < 3.5µm.** • 悬浮剂(SC): 纳米级氧化锆珠超细研磨，严格抑制奥氏熟化结晶，磨腔温度 < 35°C，粒度 D90 < 3.5µm。 **• Water Dispersible Granules (WDG): Low-temperature fluidized drying (bed temp < 48°C) to prevent melting of low-melting-point triazoles.** • 水分散粒剂(WDG): 低温流化床分段干燥（物料床温 < 48°C），防止三唑类低熔点原药热熔结块与失活。 **• Seed Treatment (FS): Color vibrancy, seed coating uniformity, film-forming speed, dusting off rate < 1.5%.** • 种子处理悬浮剂(FS): 警戒色牢度、种子包衣均匀性、成膜附着时间及落粉率（< 1.5%）。 **• Emulsion in Water (EW): High-pressure homogenization, droplet size D90 < 1.0µm, no phase separation.** • 水乳剂(EW): 高压微射流均质乳化，微滴粒径 D90 < 1.0µm，离心无分层。	• Laser Particle Size Analyzer • Heubach Seed Dustmeter • Brookfield Viscometer • Online Real-time Refractometer	CCP In-Spec 偏离工艺立即停线纠偏
FQC / OQC **成品终检与放行**	**• Dual/Triple Combination Assay: Baseline chromatographic separation of combination active ingredients (e.g., Azoxystrobin + Difenoconazole).** • 复配制剂有效成分定量分析: 色谱基线完全分离，严格符合 FAO/企业标准标称含量允差。 **• Colloid & Dispersion Physics: Suspensibility ≥ 92%, Spontaneous dispersion, Wet sieving (75µm sieve ≥ 99.0%).** • 胶体与分散物理特性: 悬浮率 ≥ 92%、水中自发分散性、湿筛通过率 ≥ 99.0%、起泡量 ≤ 30mL。 **• Dual Temperature Stress Validation: 54±2°C (14 days) accelerated heat storage + 0±2°C (7 days) freeze-thaw stability.** • 双温耐受稳定性测试: 54°C 热储 14 天原药分解率 ≤ 3% 及 0°C 低温冷储 7 天无析晶、无絮凝。	• CIPAC MT 184 / MT 174 • CIPAC MT 46.3 / MT 39.3 • Reverse-Phase HPLC-DAD	COA Approved QA 总监签章准予出厂
Traceability & Retention **留样与质量追溯**	**• Retained samples archived under 20±2°C controlled conditions for product shelf-life + 1 year (min. 3 years).** • 生产批次留样归档于 20±2°C 恒温留样库，留存至保质期后 1 年（至少 3 年）。 **• Two-dimensional serialized QR codes tracking production lot, reactor vessel, operator, and pallet logistics.** • 二维码数字化赋码，实现原药原料批号、调配釜号、操作质检员及发货物流流向双向贯通追溯。	• Constant Climate Vault • Cloud Traceability System	Archived 建立完整批生产质量档案

2. FORMULATION-SPECIFIC QC STANDARDS (FAO/CIPAC COMPLIANT) / 主要杀菌剂剂型质控标准

Formulation / 剂型	Representative Products / 代表品种	Critical Quality Attributes (CQA) / 核心理化质控指标	Key Test Method / 对应方法
SC (Suspension Conc.) **悬浮剂**	• Azoxystrobin 250g/L SC • Azoxystrobin 200g/L + Difenoconazole	• Active Content: Nominal ± 5.0% • Particle Size: D50 ≤ 1.8µm, D90 ≤ 3.5µm	• CIPAC MT 184 • CIPAC MT 148.1 • Laser Diffraction

Formulation / 剂型	Representative Products / 代表品种	Critical Quality Attributes (CQA) / 核心理化质控指标	Key Test Method / 对应方法
	125g/L SC • Pyraclostrobin 250g/L SC	• Suspensibility: ≥ 93.0% (after 54°C heat test) • Pourability Residue: ≤ 3.0%, Rinsed ≤ 0.4%	
WDG (Granules) **水分散粒剂**	• Pyraclostrobin 12.8% + Boscalid 25.2% WDG • Kresoxim-methyl 50% WDG • Mancozeb 75% WDG	• Disintegration Time: ≤ 60 seconds • Suspensibility: ≥ 88.0% (CIPAC Water D) • Residual Moisture: ≤ 1.5% • Attrition Resistance: ≥ 96.0% (No dust)	• CIPAC MT 174 • CIPAC MT 185 • CIPAC MT 178.2
FS (Flowable Seed Treatment) **种子处理悬浮剂**	• Metalaxyl-M 10g/L + Fludioxonil 25g/L + Azoxystrobin 15g/L FS • Tebuconazole 60g/L FS	• Seed Coating Uniformity: ≥ 95.0% • Dusting-off Rate: ≤ 1.5 mg/100g seed • Viscosity: 300 – 800 mPa·s (flowability) • Dye Intensity & Fastness: Uniform red/blue	• CIPAC MT 175 • Heubach Method • CIPAC MT 192
EW / ME (Emulsion) **水乳剂 / 微乳剂**	• Difenoconazole 250g/L EC/EW • Tebuconazole 25% EW	• Emulsion Droplet Size: D90 ≤ 0.8 µm • Thermal Stability (54°C, 14d): No creaming / oiling • Cold Stability (0°C, 7d): Transparent, no separation	• CIPAC MT 36.3 • CIPAC MT 39.3 • Laser Particle Size

3. SUPPORTING EVIDENCE 1: OFFICIAL COA TEMPLATE / 佐证文件一：成品出厂检验报告单 (COA)

AGROSPEAR CROPCARE CO., LTD.

CERTIFICATE OF ANALYSIS / 产品检验分析报告单 (FUNGICIDE)

Product Name / 品名: Azoxystrobin 200g/L + Difenoconazole 125g/L SC (嘧菌酯 200g/L + 苯醚甲环唑 125g/L 悬浮剂)

Batch No. / 生产批号: AS-FUN-20260826-05

Invoice / Order No. / 订单号: AGR-EXP-2026-902

Batch Quantity / 批量: 10,000 Liters (1L x 12/Carton)

Manufacture Date / 生产日期: 2026-08-26

Expiry Date / 有效期至: 2028-08-25

Sampling Date / 取样日期: 2026-08-26

Inspection Date / 判定日期: 2026-08-27

Test Parameter / 检验项目	Specification Limit / 标准指标	Test Result / 实测结果	Standard Method / 检验方法	Verdict / 判定
Appearance / 外观性状	Off-white to ivory homogeneous suspension (类白色至象牙白均质流体)	Conforms / 符合要求	Visual Check	Pass
Active Ingredient 1: Azoxystrobin (嘧菌酯)	200.0 ± 10.0 g/L (190.0 – 210.0 g/L)	**202.4 g/L**	HPLC-DAD (Agilent 1260)	Pass
Active Ingredient 2: Difenoconazole (苯醚甲环唑)	125.0 ± 6.25 g/L (118.75 – 131.25 g/L)	**126.1 g/L**	HPLC-DAD (Agilent 1260)	Pass
pH Value (1% aqueous solution)	5.5 – 8.5	6.8	CIPAC MT 75.3	Pass
Suspensibility / 悬浮率	≥ 90.0 % (CIPAC Standard Water D)	**97.2 %**	CIPAC MT 184	Pass
Particle Size Distribution / 粒度 (D90)	≤ 3.5 µm	2.4 µm (D50: 1.1 µm)	Laser Diffraction	Pass
Wet Sieve Test / 湿筛试验 (75µm sieve)	≥ 99.0 % through	99.9 %	CIPAC MT 185	Pass
Persistent Foaming / 持久起泡性	≤ 35 mL after 1 min	10 mL	CIPAC MT 47.2	Pass
Pourability / 倾倒性 (Rinsed Residue)	Residue ≤ 3.5 %, Rinsed ≤ 0.4 %	Residue: 1.5 %, Rinsed: 0.15 %	CIPAC MT 148.1	Pass
Heat Stability (54±2°C, 14 days) / 热储	AI Degradation ≤ 3.0%, Suspensibility ≥ 90%	Azoxy Deg: 0.6%, Difen Deg: 0.8%, Susp: 96.0%	CIPAC MT 46.3	Pass
Low Temp. Stability (0±2°C, 7 days) / 冷储	No crystal formation or phase separation (无晶体析出)	Conforms / 无析出	CIPAC MT 39.3	Pass

CONCLUSION / 检验结论: PASS - All parameters conform to FAO Specification and In-house Quality Standard. Approved for release. (合格 - 各项指标符合 FAO 规格及企业内控标准，准予放行出厂。)

Analyst / 检测员:
Zhang Wei (张伟)
Date: 2026-08-27

Reviewer / 复核员:
Li Ming (李明)
Date: 2026-08-27

QA Director / QA总监:
Dr. Alex Chen (陈博士) *[Approved]*
Date: 2026-08-27

4. SUPPORTING EVIDENCE 2: BATCH RELEASE & OOS DISPOSITION / 佐证文件二：批质量放行审批与超标处理规程

Release Gate / 质量关卡	Verification Requirements / 审核项目	Status / 状态	Verified By / 审核人
1. Raw Material Gate (IQC)	Active ingredient assay & isomer ratios, toxic impurity limit (ETU/ solvents), EVOH bottle compatibility	Verified Pass	IQC Lead
2. Process Parameters (IPQC)	Bead milling temp < 35°C, fluid bed low-temp drying < 48°C, filter screen integrity, torque & volume accuracy	Verified Pass	IPQC Officer
3. Laboratory FQC (COA)	Dual-active chromatographic assay, suspensibility, wet sieving, accelerated heat/cold stability	Verified Pass	Lab Supervisor
4. Packaging & Traceability	Dual 2D QR-code serialization, laser batch code legibility, 100% inductive foil seal integrity	Verified Pass	Packaging QA

OOS (Out of Specification) & CAPA Protocol: If any batch fails specifications, the Batch Release Gate is locked automatically in the ERP. Investigation is conducted across 3 tiers (Analytical error → Formulation process deviation → Raw material drift). Reprocessing requires a formal CAPA protocol approved by the Plant General Manager and QA Director.

超标调查（OOS）与纠正预防措施（CAPA）：凡任一指标出现偏差，ERP 系统自动锁定该批产品出厂闸口并张贴物理红卡。启动三级溯源调查（实验室分析误差 → 生产工艺偏离 → 原料批次漂移）。返工或重组方案必须经工厂总经理与质量总监双重书面签批方可执行。